CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Cardboard Arcade

Goal: Build a small tabletop cardboard skill game with one clear goal, a reliable mechanism, and a fast reset.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 30

Ask | Plan | Build | Test | Improve

You are done when

Let a partner play, show the ten-second reset, and name the revision that improved the experience.

You will use

- Cardboard
- Paper cups
- Craft sticks
- Paper fasteners
- String
- Tape
- Markers
- Marbles or contained game tokens
- Score cards
- Tabletop game cards
- Play trays

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one tabletop game card.
- Place tokens inside the play tray.

2 Step 2: Game safety — 3 min

- Keep tokens inside the table boundary.
- Stop any part that pinches.
- Use no launcher.

3 Step 3: Plan the game — 4 min

- Write the game goal in one sentence.
- Sketch the token path.

4 Step 4: Build the game — 9 min

- Build the smallest playable version.
- Add one reliable cardboard connection.
- Make a clear start point.

5 Step 5: Play test — 7 min

- Ask a partner to play without help.
- Watch the first confusing moment.
- Record one unreliable part.

6 Step 6: Improve the reset — 4 min

- Change the confusing part.
- Reset the game twice.

7 Step 7: Reset the arcade — 2 min

- Count every token.
- Place the game on the storage shelf.

8 Step 8: Exit check — 3 min

- State the goal.
- Show how the game resets.

What to turn in

A playable game with a ten-second reset.

No computer? Another way to do it

Make a paper board game with the same rules.

Finished early? Next challenge

Level Two: Add one optional rule that makes the same game more challenging without rebuilding it.

Stay safe

Keep every moving token inside the tabletop play area, use no launchers, and stop any mechanism that pinches or sends parts toward people.

Grade targets

Grade 5: One goal and one mechanism.
Grade 6: Add a scoring rule and a ten-second reset.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.